

OOP Lab sample Oral questions:

Section A: Basic Java & OOP Concepts

1. What is Object-Oriented Programming (OOP)?
2. What are the main principles of OOP?
3. What is a class and an object in Java?
4. What is the difference between a constructor and a method?
5. What is constructor overloading?
6. What is method overloading?
7. What is the role of the this keyword in Java?
8. What is the difference between static and non-static members?
9. What are access modifiers in Java?
10. What is encapsulation?

Section B: Inheritance, Polymorphism & Abstraction

11. What is inheritance? Explain its types.
12. What is single inheritance?
13. What is method overriding?
14. What is runtime polymorphism?
15. What is compile-time polymorphism?
16. What is an abstract class?
17. What is an interface in Java?
18. Difference between abstract class and interface?
19. Can a class implement multiple interfaces? Explain.
20. What is dynamic method dispatch?

Section C: Exception Handling

21. What is exception handling in Java?
22. What is the difference between checked and unchecked exceptions?
23. What is the purpose of try-catch-finally block?
24. What happens if an exception is not handled?
25. What is the use of throw and throws keywords?
26. What is the difference between throw and throws with an example?
27. How would you handle multiple exceptions in a single program?
28. Why is finally block important? Can it be skipped?

Section D: Multithreading

29. What is a thread in Java?
30. Difference between process and thread?
31. What is thread lifecycle?
32. What is synchronization in Java?

33. What is the use of sleep() and join() methods?
34. What steps are involved in connecting Java to a database?
35. Why are PreparedStatement preferred over Statement?
36. How do you handle transactions in JDBC?
37. What is SQL injection and how can it be prevented?